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Title: Chemical Spill Cleanup Procedure

Chemical Spill Clean Up Procedure

This document should be read in conjunction with Chemical Safety procedure.

1. Warn other staff in the near vicinity of a spillage
2. Be aware of the dangers of the chemicals you are using. Refer to COSHH / SDS in the Hazard Manual for each Department. These will give a list of the safety information and first aid measures to be taken for all chemicals used in the department.
3. Collect spill kit. Each kit is contained in a white plastic bucket which has the following items:

Plastic apron b. Face shield c. Nitrite gloves d. Spillex™ Products e. Plastic bags f. brush and shovel g. pH paper 0-14 h. Presept™ granules (biological spills)

Procedure:

a. When a spill has occurred, first decide whether you know what the chemical involved is or not. Call the safety officer or supervisor.
b. Consult material data safety sheets (COSHH) available in the hazard register. These will alert you to the dangers and tell you what precautions you need to take.
c. Be aware that the neutralization process maybe be exothermic and /or produce hazardous fumes. Call safety officer.
d. Safety equipment must be worn during the cleanup process
e. All spills must be neutralized with the appropriate Spillex™ product
    1. Spillex A Used for acids
    2. Spillex C Used for alkalis / caustic material
    3. Spillex XS Used for solvents
    4. Spillex FP Used for Formalin

Starting from the outside of the spill, pour the Spillex™ slowly around the spill in decreasing circles until all the liquid is absorbed and the pH is close to 7.0. Check with pH paper provided.

Sweep up absorbed liquid and place the slurry into the plastic bags provided. Plastic this into another yellow biohazard bag, together with gloves and apron, for disposal.

For spills of greater than 1 litre of any of the following or combination thereof, **CALL 9777** alert the Hospital Fire Officers and evacuate the area.

Concentrated Nitric acid

Concentrated Sulphuric acid

Concentrated Diethyl Ether

Concentrated Formaldehyde

Concentrated Phosphoric acid

Ammonia

Chloroform

THE OSHA LABORATORY STANDARD

The basis for this standard (29 CFR 1910.1450) is a determination by the Occupational Safety and Health Administration (OSHA), after careful review of the complete rule-making record, that laboratories typically differ from industrial operations in their use and handling of hazardous chemicals and that a different approach than that found in OSHA's substance specific health standards is warranted to protect workers. The final standard applies to all laboratories that use hazardous chemicals in accordance with the definitions of laboratory use and laboratory scale provided in the standard. Generally, where this standard applies it supercedes the provisions of all other standards in 29 CFR, part 1910, subpart Z, except in specific instances identified by this standard. For laboratories covered by this standard, the obligation to maintain employee exposures at or below the permissible exposure limits (PELs) specified in 29 CFR, part 1910, subpart Z is retained. However, the manner in which this obligation is achieved will be determined by each employer through the formulation and implementation of a Chemical Hygiene Plan (CHP). The CHP must include the necessary work practices, procedures and policies to ensure that employees are protected from all potentially hazardous chemicals used or stored in their work area. Hazardous chemicals as defined by the final standard include not only chemicals regulated in 29 CFR part 1910, subpart Z, but also any chemical meeting the definition of hazardous chemical with respect to health hazards as defined in OSHA's Hazard Communication Standard, 29 CFR 1910.1200(c).

Among other requirements, the final standard provides for employee training and information, medical consultation and examination, hazard identification, respirator use and record keeping. To the extent possible, the standard allows a large measure of flexibility in compliance methods.

EMPLOYEE RIGHTS AND RESPONSIBILITIES

Employees have the right to be informed about the known physical and health hazards of the chemical substances in their work areas and to be properly trained to work safely with these substances.

Employees have the responsibility to attend training seminars on the Laboratory Standard and Chemical Hygiene Plan and to stay informed about the chemicals used in their work areas. They have the responsibility to use safe work practices and protective equipment required for safe performance of their job. Finally they have the responsibility to inform their supervisors of accidents and conditions or work practices they believe to be a hazard to their health or to the health of others.

HAZARDOUS CHEMICALS

The Laboratory Standard defines a hazardous chemical as any element, chemical compound, or mixture of elements and/or compounds which is a physical or health hazard.

A chemical is a **physical hazard** if there is scientifically valid evidence that it is a flammable, a combustible liquid, a compressed gas, an explosive, an organic peroxide, an oxidizer, pyrophoric, unstable material (reactive), or water-reactive.

A chemical is a **health hazard** if there is statistically significant evidence based on at least one study conducted in accordance with established scientific principles that acute or chronic health effects may occur in exposed employees. Included are:

- carcinogens
- irritants
- reproductive toxins
- corrosives
- sensitizers
- radioactive material

- hepatotoxins (liver)
- nephrotoxins (kidney)
- agents that act on the hematopoietic system (blood)
- agents that damage the lungs, skin, eyes, or mucous membranes

See Appendix I, Glossary, for definitions of these terms.

In most cases, the label will indicate if the chemical is hazardous. Look for key words like **caution, hazardous, toxic, dangerous, corrosive, irritant, carcinogen**, etc. Old containers of hazardous chemicals (before 1985) may not contain hazard warnings.

If you are not sure a chemical you are using is hazardous, review the **Safety Data Sheet (SDS)** or contact your supervisor, instructor, or the QA office.

Designated areas must be established and posted for work with certain chemicals and mixtures, which include **select carcinogens**, **reproductive toxins**, and/or substances which have a **high degree of acute toxicity**. A designated area may be the entire laboratory, an area of a laboratory or a device such as a laboratory hood. Designated area stickers are available from QA.

SAFETY DATA SHEETS (SDSs)

A Safety Data Sheet (SDS) is a document containing chemical hazard and safe handling information prepared in accordance with the OSHA Hazard Communication Standard. A sample SDS is included at the end of Part I.

Chemical manufacturers and distributors must provide a SDS the first time a hazardous chemical/product is shipped to a facility. Only SDSs received must be retained and made available to laboratory workers. However, you can request an SDS for any laboratory chemical from the manufacturer or distributor.

CHEMICAL INVENTORIES

The OSHA Laboratory Standard does not require chemical inventories; however, it is prudent to adopt this practice. An annual inventory can reduce the number of unknowns and the tendency to stockpile chemicals.

ROYAL HOSPITAL PATHOLOGY - CHEMICAL HYGIENE PLAN

This document serves as the written Chemical Hygiene Plan (CHP) for Pathology laboratories using chemicals at Royal Hospital. The CHP is a regular, continuing effort, not a standby or short term activity. Departments, divisions, sections, or other work units engaged in laboratory work whose hazards are not sufficiently covered in this written manual must customize it by adding their own sections as appropriate (e.g. standard operating procedures, emergency procedures, identifying activities requiring prior **approval before commencing work.**

RELATED DOCUMENTS

HSE Induction – New Staff

HSE Audit template

Technical Procedure Risk assessment

COSHH risk assessment template

Waste Management Audit checklist

Blood borne Pathogen Control Plan

Chemical Hygiene Plan

Pathology (Initial Release 2016)

Checked By: Emergency & Disaster Preparedness Committee
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Purpose of policy

To create a safe environment free from chemical waste hazards and protect staff, patients, and the public from these impacts. Moreover to ensure health safety precautions of hazardous chemical waste when dealing, storing, transporting, and disposing of chemical waste.

Scope

This guideline is applicable to all healthcare workers, cleaning staff, and other workers including contracted service workers in healthcare facilities who deal with chemical waste while, storing, transporting, and safely discarding it in the royal hospital.

Definition

Hazardous chemical waste: Any excess, unusable and expired chemical that can cause an effect on human health and on the environment. As well, as any Hazardous substances that become waste and are classified as one of the following toxic, carcinogenic, harmful, cause-specific organ damage, irritant, Respiratory sensitizing, corrosive, or Toxic for reproduction.

Roles & Responsibilities

A. All Patient Care Staff

- All staff should comply with this Guideline and report any hazards or incidents immediately to their supervisors and HSE.
- Properly handle and store & transport chemicals.
- Should be aware of the risks posed by handling and storing
- Never place chemical bottles & containers at office
- Keep filled containers in an area that excludes the risk of injury to the staff and visitors.

Focal points:

- Inform and update the quantity of hazardous chemical waste in each section.
- Report any chemical spillage and ensure a proper spillage kit is in place (depending on the chemical type they use in this area).
- Safety Data Sheet must be consulted for appropriate safety methods
- Ensure that waste must be properly labeled, stored in a suitable container, and housed appropriately until collection.

B. Hospital management:

- Provide all required resources for the implementation of this Guideline

C. Departmental Management

- Enforce this guideline
- Education and training of staff on safe disposal of hazardous chemical waste.

D. Health, Safety, and Environment Technician

- Monitor compliance with Guidelines through regular audits.
- Provide education and training regarding handling and safe disposal of chemicals waste.
- Regular inspection to ensure safe disposal of hazardous chemical waste
- pointed focal points in each section (dealing with hazardous chemicals or generated).
- Ensure arrangements must be in place for waste to be collected regularly by the licensed waste contractor for appropriate off-site treatment.

E. Housekeeping Staff:

- Adhere to this policy.
- Should not do any action without direct instruction from the focal point or the technician.
- Do not refill or move the bottles or containers without PPE
- Immediately report incidents of spillage, skin contact, or breathing problems.

Container labeling:

All chemical waste containers must be clearly labeled to accurately identify their contents. These labels are available in this document and can be printed. The following details must be included on the waste label prior to the chemical waste being taken to your local Stores.

Each Container must have a **label** that identifies the

1. Full name + quantity of the chemical waste
2. Date of generation
3. The tag or lapel must be mentioned (chemical classification)
   (See appendix3)

Hazardous waste storage requirements:

- All waste must be stored in containers,
- Containers must be kept closed at all times
- except when adding or removing waste.
- Containers must be labeled or clearly marked with words that describe the contents of the waste and the words "Hazardous Waste"
- Containers must be in good shape and not leaking and must be compatible with the waste they contain.
- Containers must be stored at or near the point of generation and under the control of the generator of the waste.
- Containers must be segregated by chemical compatibility during storage.
- Well-ventilated.
- Liquid and solid materials are stored separately where possible, to avoid contamination in case of spillage.
- Spill Control and clean-up materials must be available.

Specification of safety cabinet:

- Designed for the safe storage of chemicals and flammable liquids
- Storage capacity (as per requirement)
- Fire-resistant, leak-tested design
- Manual close (better self-closing)
- Warning Label
- Complies with OSHA, NFPA standard

For more details about the specifications of the safety, cabinet see Appendix 4.

Collection of waste:

1. Contact with an authorized company
2. Pointed the collection areas (To reduce the risk of contact and spillage).
3. Only authorized persons must deal with waste while collecting
4. Use the proper PPE.
5. Use leak-proof carts that are readily cleanable to transport the chemical waste from the point of generation or storage to the point of disposal and transport.
6. Decontaminate carts used for transporting waste within the hospital

Empty chemical bottles:

- The staff have insured that bottles are empty without any chemicals even if they drop
- Storing empty bottles in a safe place to avoid breaking
- Keep the empty bottles in the carton box with clear labels
- Can reuse the empty bottles and insure its clear of any chemicals
- Dispose of the empty chemical bottles in double black bags after that put them in boxes with clear labels (chemical glass bottles) to send for final disposal as general waste.
- Clean and wash the bottles before reuse
- Do not dispose of the empty chemical bottles in the yellow bags
-

Transport

On-site transport

On-site transport of waste is usually from the initial storage point to assembly storage.

Trolleys used for the chemicals wastes movement must be designed to prevent leakage, and easily cleaned.

Off-site transport

- The authorized company must have approval from environmental authorities to practice transporting and disposing of hazardous chemical materials.

- The authorized contractor must comply with Health, safety, and environmental condition for proper transportation inside the hospital.

  -Wear proper PPE
  -Spillage materials must be available while collection.
  -Proper Trolley for transport.

Disposal of waste:
-Proper chemical waste containers are required to dispose of in accordance with the Health, Safety, and Environment Guideline and authorized Company procedures for the last disposal.

-Final disposal going for treatment

Injuries related to improper disposal and handling can occur for the following reasons:

- Inappropriate usage and handling practices during the provision of care by a healthcare professional.
- Improper design storage.
- Overfilling the container.

Occupational health and safety:

The hospital must ensure that hazardous chemical waste is handled, disposed and transported safely. The waste management plan and procedures should be readily available to all workers involved.

All personnel who handle or deal with chemicals must have access to information about:

- SDS Safety Data Sheet must be available
- Occupational hazards of unprotected exposure
- Policies and procedures for specific hazardous chemicals waste handling & transporting and prevention of injury and occupational disease.
- Personal protective equipment.
- Access to the Incident Report system IRS and report in case of exposure to hazardous chemical waste.
- Availability and access to first aid resources.

Training

Ensuring staff and students are provided with suitable instruction and training to enable them to work with hazardous substances safely. Instruction should include

- Information on the substances used
- Risks to health presented by the use of those substances
- Relevant workplace exposure limits.
- Relevant safety data information
- The significant findings of chemical risk assessments
- Precautions to take to prevent or reduce exposure
- Correct operation and use of equipment and control measures.
- Correct safe disposal.

Security

- Access and supervisory arrangements for the use and storage of hazardous chemicals are in place and are relevant to the risks associated with the chemicals used.
- Where regulated substances are in use, documented procedures are in place to ensure suitable access control.

Monitoring performance:

- An audit should be performed semiannually to determine current performance in terms of safety, efficiency, environmental impact assessment, costs, and regulatory compliance

-All incidents and adverse events related to hazardous waste materials should be recorded. This is essential for identifying the factors that cause hazards while handling, storage, and transporting.

- Determining the effectiveness of the waste management plan in reducing the incidence and severity of injuries.

Appendix(1)

Request form
hazardous waste disposal

Department:	Focal point person: Signature :		comment
Date of collection:	Supervise by : Signature:	Office Number	Mobile

Name of chemical	Type of Chemical	Location	Type of container	Weights KG /litter

Appendix2

Pickup Form

Department:	**Focal point person:** **Signature :**	**phone**	
Date of collection:	Supervise by : Signature:	Office Number	**Mobile**

Name of chemical	Type of Chemical	Location	Weights KG /litter

Appendix3

HAZARDOUS WASTE LABEL

1. Collect waste in a screw cap container and keep the container closed on except level.
2. Put on and fill in a hazardous waste label when you start collecting the waste.
3. Keep the empty bottles in the carton box with clear labels
4. Do not accumulate waste bottles in the laboratory. Contact directly with HSE after filling out the waste collection template. (Note: Arranged for quarterly discarding)

CHEMICAL CONTENT HAZARD

_____________________________ ☐ TOXIC

_____________________________ ☐ FLAMMABLE

___________________________ ☐ OXIDIZER

___________________________ ☐ CORROSIVE

_____________________________ ☐ REACTIVE _____(Describe

____ Date Full

Add to the label as necessary as more waste is added:

A. A generic waste name (e.g. chlorinated solvents) – optional

B. Each chemical present – required (no chemical formulas, structures or abbreviations)

C. The hazard of the waste (flammable, oxidizer, corrosive, toxic, reactive) –required --If the waste is not flammable (flash point < 2 or > 12.5) or reactive, check toxic.

D. When the container is full, write in the date (month, day, and year).

Appendix4

Flammable chemical storage cabinet

Safety specification of chemical storage cabinet:

- Designed for the safe storage of chemicals and flammable liquids
- Storage capacity(as per requirement)
- Fire-resistant, leak-tested design
- Manual close(better self closing)
- Warning Label
- Complies with OSHA, NFPA standard

Request a safety cabinet as per your needs and determine the following:

- **Cabinet body:** Double fireproof steel construction
- **Color**: Easily identify, organize and segregate liquids by using the correct color cabinets.
- **Size and type of** storage cans: Determine the length and width
- **Capacity**: Select specialty cabinets for on-the-spot needs larger cabinets for expanded or large quantity storage.
- **Numbers of door**
- **Total Number of Shelves**

References

Title of book/ journal/ articles/ Website	Author	Year of publication	Page
WWW.OSHA.GOV	Occupational safety and health administration-U.S. department of labour	2011	
http://www.mass.gov/dep/service/regulations/310cmr30.pdf	Hazardous waste identification	2008	

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